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2024

Answer No. 11

+ The scope of statistics is vast and extends across various fields, providing tools and methods for collecting, analyzing, interpreting and presenting data. Here's a brief overview of its scope:

a. Business and Economics:
b. Science and Engineering:
c. Social sciences
d. Sports and entertainment
e. Agriculture

+ While statistics is a powerful tool for analyzing and interpreting data, it has certain limitations. These limitations are arised from the nature of data, the methods used and the interpretation of results. Below is a detailed discussion of the limitation of statistics:

a. Statistics deals with aggregates, not individuals.
b. Statistics cannot study qualitative phenomena
c. Dependence on Quality of data
d. Ethical Concerns
e. Time and cost constraints

Answer No. 12

Statistics is a branch of mathematics that deals with the collection, organization, analysis, interpretation and presentation of data.

Class	f	m	fm	fm^2
0-20	12	10	120	1200
20-40	16	30	480	14400
40-60	35	50	1750	87500
60-80	24	70	1680	117600
80-100	13	90	1170	105300
	$N=100$		$\Sigma fm = 5200$	$\Sigma fm^2 = 326000$

$$\text{Mean}(\bar{x}) = \frac{\Sigma fm}{N} = \frac{5200}{100} = 52$$

class	f	cf
0-20	12	12
20-40	16	28
40-60	35	63
60-80	24	87
80-100	13	100
	$N=100$	

$$\text{Position of median}(Md) = \left(\frac{N}{2}\right)^{th}\text{class}$$

$$= \left(\frac{100}{2}\right)^{th}\text{class}$$

$$= 50^{th}\text{ class}$$

Here, cf just greater than 50 is 63. So, it's corresponding class is (40-60)

$l = 40$, $f = 35$, $\frac{N}{2} = 50$, $h = 20$, $cf = 28$

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$$\text{median} = l + \frac{\frac{N}{2} - cf}{f} \times h$$

$$= 40 + \frac{50 - 28}{35} \times 20$$

$$= 40 + 52.57$$

$$S.D(\sigma) = \sqrt{\frac{\Sigma fm^2}{N} - \left(\frac{\Sigma fm}{N}\right)^2}$$

$$= \sqrt{\frac{326000}{100} - \left(\frac{5200}{100}\right)^2}$$

$$= \sqrt{3260 - 2704}$$

$$= \sqrt{556} = 23.57$$

Answer No. 13

Correlation refers to the statistical relationship or association between two variables. It measures the degree to which changes in one variable are associated with changes in another variable. It ranges from -1 (perfect negative correlation) to +1 (perfect positive correlation) with 0 indicating no correlation.

We use the formula:

$$r = \frac{n\Sigma xy - (\Sigma x)(\Sigma y)}{\sqrt{[n\Sigma x^2 - (\Sigma x)^2][n\Sigma y^2 - (\Sigma y)^2]}}$$

$n = 20$

588.65625

x	y	x^2	y^2	xy
55	65	3025	4225	3575
70	40	4900	1600	2800
40	30	1600	900	1200
30	50	900	2500	1500
90	60	8100	3600	5400
80	70	6400	4900	5600
60	50	3600	2500	3000
80	50	6400	2500	4000
90	60	8100	3600	5400
80	70	6400	4900	5600

$\Sigma x = 675$ $\Sigma y = 545$ $\Sigma x^2 = 49425$ $\Sigma y^2 = 31225$ $\Sigma xy = 38075$

Putting values:

$$r = \frac{10(38075) - (675)(545)}{\sqrt{[10(49425) - (675)^2][10(31225) - (545)^2]}}$$

$$\text{or, } r = \frac{380750 - 367875}{\sqrt{(494250 - 455625)(312250 - 297025)}}$$

$$\text{or, } r = \frac{12875}{\sqrt{(38625)(15225)}}$$

$$\text{or, } r = \frac{12875}{\sqrt{588065625}}$$

$$\text{or, } r = \frac{12875}{24250}$$

or, $r \simeq 0.53$

Interpretation:
A correlation coefficient of 0.53 indicates a moderate positive linear relationship between maths and basic statistics scores. As maths scores increase, Basic statistics scores tend to increase as well moderately.

Answer No. 14

Regression is a statistical method used to model the relationship between a dependent variable and one or more independent variables.

x	y	x^2	y^2	xy
2	10	4	100	20
4	15	16	225	60
6	22	36	484	132
8	32	64	1024	256
10	46	100	2116	460
$\Sigma x = 30$	$\Sigma y = 125$	$\Sigma x^2 = 220$	$\Sigma y^2 = 3949$	$\Sigma xy = 928$

i) Let regression equation of y on x

$$y = a + bx \quad - (i)$$

Where,

y = dependent variable
x = independent variable
a = y-intercept
b = regression coefficient

10:30

Here, a and b are constants and need to determine by solving following linear equations obtained from method of least square.

$\Sigma y = na + b\Sigma x$

or, $125 = 5a + b\,30$

or, $5a + 30b = 125$ ... (ii)

$\Sigma xy = a\Sigma x + b\Sigma x^2$

or, $928 = a\,30 + b\,220$

or, $30a + 220b = 928$ ... (iii)

Multiplying eqn (ii) by 30 and eqn (iii) by 5 and subtract eqn (iii) from (ii)

$150a + 900b = 3750$

$-\ 150a \oplus 1100b = 4640$

$-200b = -890$

or, $b = \frac{890}{200} = 4.45$

from eqn (ii)

$5a + 30b = 125$

or, $5a + 30 \times 4.45 = 125$

or, $5a = -8.5$

or, $a = \frac{-8.5}{5}$

$\therefore a = -1.7$

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Hence required regression eqn is

$\hat{y} = -1.7 + 4.45X$

ii. $\hat{y} = -1.7 + 4.45(10) = -1.7 + 44.5$

$= 42.8$

so, the estimated maintenance cost is Rs 42,800

iii. The slope is 4.45, which means;

"For every one additional year in the age of the computer the annual maintenance cost increases by approximately Rs. 4450. or

For each additional year of age, the annual maintenance cost increases by about Rs 4,450

Answer No. 15

A random variable X is said to follow Poisson distribution if it takes non-negative values and its probability mass function is,

$$P(X=x) = \frac{e^{-\lambda}\lambda^x}{x!} \quad x = 0, 1, 2 \ldots\ldots$$

We are given;

probability that a lens is defective : $p = \frac{1}{500} = 0.002$

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$\frac{10!}{2!(8)!}$ $\binom{10}{2}$ $^nC_x = \frac{n!}{x!(n-x)!}$ $10! \binom{10}{1} = \frac{10!}{1!(9)!}$

- Number of lenses in a packet : $n = 10$
- The probability of a lens being non-defective $q = 1 - p = 0.998$

This is a binomial distribution problem, where we define:

$$X \sim Binomial\ (n = 10,\ p = 0.002)$$

The binomial probability mass function is:

$$P(X = x) = \begin{cases} ^nC_x\, p^x\, q^{n-x}, & x = 0, 1, 2, \ldots n \\ 0 & \text{otherwise} \end{cases}$$

i) $P(X = 0) = {}^{10}C_0\ (0.002)^0 \cdot (0.998)^{10}$

$= 1 \cdot 1 \cdot (0.998)^{10}$

$= \approx 0.9802$

ii) $P(X \geq 1) = 1 - P(X = 0)$

$= 1 - 0.9802$

$= 0.0198$

iii) $P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$

$= 0.9802 + {}^{10}C_1\ (0.002)^1 (0.998)^9$

$+ {}^{10}C_2\ (0.002)^2 (0.998)^8$

$= 0.9802 + 10 \cdot 0.002 \cdot (0.998)^9 + 45 \cdot (0.002)^2 \cdot (0.998)^8$

$= 0.9802 + 0.0198 + 0.000177 \approx 0.99997$

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Answer No. 16

We are given the following:

- Confidence level: 95%
- Margin of Error (E): 0.25 minutes
- σ = 1.40 minutes.
- Z = Z-score for the desired confidence
- level = 1.96

Sample size formula:

$$n = \left(\frac{Z \cdot \sigma}{E}\right)^2$$

$$\text{or, } n = \left(\frac{1.96 \cdot 1.40}{0.25}\right)^2$$

$$\text{or, } n = \left(\frac{2.744}{0.25}\right)^2$$

$$\simeq 120.45$$

Since sample size must be a whole number;

$n = 121$

So, the dean will need to take a sample of at least 121 students to estimate the average time with 95% confidence and a maximum error of 0.25 minute.

Answer No. 17

Sampling is the process of selecting a subset from a larger group (called the population) in order to make

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inference or draw conclusions about the population as a whole.

Given population:

$\{2, 8, 14, 20\}$

i) The total number of such samples is:

$$\binom{4}{2} = 6$$

So, the possible samples of size 2 are:

Sample no.	Sample
1	(2, 8)
2	(2, 14)
3	(2, 20)
4	(8, 14)
5	(8, 20)
6	(14, 20)

ii) Step 1: Compute the population mean

$$\mu = \frac{2+8+14+20}{4} = \frac{44}{4} = 11$$

Step 2: Compute mean of each sample

Sample	Sample mean
(2, 8)	$(2+8)/2 = 5$
(2, 14)	$(2+14)/2 = 8$
(2, 20)	$(2+20)/2 = 11$
(8, 14)	11
(8, 20)	14
(14, 20)	17

$\frac{24}{6015}$
$\frac{241602.5}{6015}$

Step 3: Compute the expected value (mean of the sample means)

Average of sample means $= \frac{5+8+11+11+14+17}{6}$

$= 11$

This equals the population mean $\mu = 11$

Since, the average of all possible sample means equals the population mean, we conclude that:
"The sample mean is an unbiased estimator of the population mean."

Group C

For ~~Group~~ Company A

Time (in Second)	f	m	fm	fm^2
0-2	5	1	5	5
2-4	16	3	48	144
4-6	13	5	65	325
6-8	7	7	49	343
8-10	5	9	45	405
10-12	4	11	44	484
	$N = 50$		$\Sigma fm = 256$	$\Sigma fm^2 = 1706$

$$Mean(\bar{x}) = \frac{\Sigma fm}{N} = \frac{256}{50} = 5.12 \text{ seconds}$$

Now,

65536

91864

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$$S.D(\sigma) = \sqrt{\frac{\Sigma fm^2}{N} - \left(\frac{\Sigma fm}{N}\right)^2}$$

$$= \sqrt{\frac{1706}{50} - \left(\frac{256}{50}\right)^2}$$

$$= \sqrt{34.12 - 26.21}$$

$$= \sqrt{7.91}$$

$$= 2.81$$

$$CV = \frac{sd}{mean} \times 100\%$$

$$= \frac{2.81}{5.12} \times 100\%$$

$$= 54.88\%$$

For ~~Group~~ Company B

class	f	m	fm	fm^2
0-2	2	1	2	2
2-4	7	3	21	63
4-6	12	5	60	300
6-8	19	7	133	931
8-10	9	9	81	729
10-12	1	11	11	121
	N = 50		$\Sigma fm = 308$	$\Sigma fm^2 = 2146$

$$Mean(\bar{x}) = \frac{\Sigma fm}{N} = \frac{308}{50} = 6.16$$ seconds

$$SD(\sigma) = \sqrt{\frac{\Sigma fm^2}{N} - \left(\frac{\Sigma fm}{N}\right)^2}$$

$$= \sqrt{\frac{2146}{50} - \left(\frac{308}{50}\right)^2}$$

$\sqrt{42.92 - 37.94}$
$\sqrt{4.98}$
2.23

$CV = \frac{sd}{mean} \times 100\%$
$= \frac{2.23}{6.16} \times 100\%$
$= 36.20\%$

Since CV of group B is less than CV of group A so, Group B's or Company B's computers are more consistent.

Answer No. 19

A continuous random variable X is said to have a normal distribution with parameters μ and σ if it assumes all real values within the interval $-\infty$ to $+\infty$ and its probability density function is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad -\infty < x < \infty$$

The standard normal distribution is a special case of the normal distribution where the mean is 0 and the sd is 1.

$$Z \sim N(0, 1)$$

mean(μ) = 5000
sd (σ) = 1000
If X follows normal distribution then standard normal variate of X is,

$$Z = \frac{X - \mu}{\sigma}$$

$$Z = \frac{X - 5000}{1000}$$

a) $P(X < 5012)$
when $X = 5012$

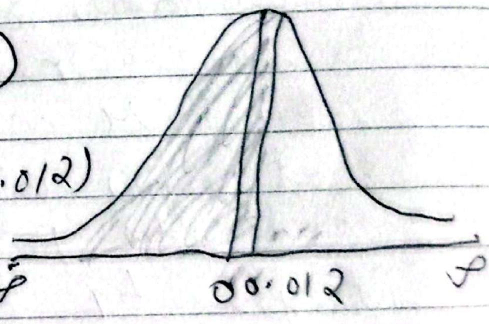
$$Z = \frac{5012 - 5000}{1000}$$

$$= 0.012$$

$$P(X < 5012) = P(Z < 0.012)$$

$$\therefore P(Z < 0.012)$$

$$= P(-\infty < Z < 0) + P(0 < Z < 0.012)$$
$$= 0.5 + 0.0040$$
$$= 0.504$$



b) $P(4000 < X < 6000)$
when $X = 4000$

$$Z = \frac{4000 - 5000}{1000}$$

$$= -1$$

when $X = 6000$

$$Z = \frac{6000 - 5000}{1000} = 1$$

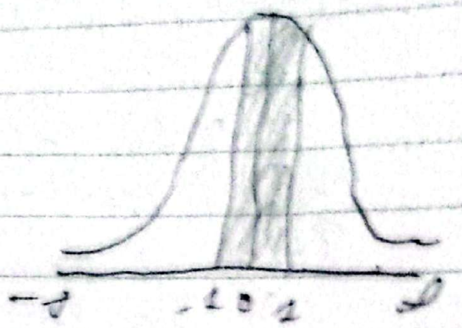
$P(4000 < X < 6000) = P(-1<z<1)$

or $P(z\ -1< z<1)$

$= P(-1<z<0) + P(0<z<1)$

$= 0.3413 + 0.3413$

$= 0.6826$



c) $P(X>7000)$

When $X = 7000$

$$Z = \frac{7000-5000}{1000}$$

$= 2$

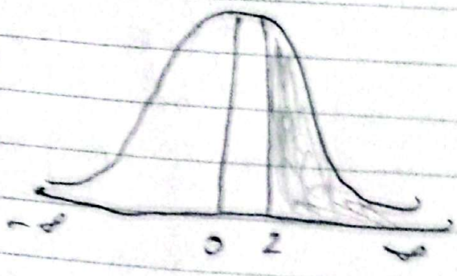
$P(X>7000) = P(z>2)$

$P(z>2)$

$= P(0<z<\infty) - P(0<z<2)$

$= 0.5 - 0.4772$

$= 0.0228$



Answer No. 2

Design of Experiment is a systematic approach to determining the relationship between factors affecting a process and the output of that process.

Properties of design of experiment

a. Replication

b. Randomization

c. Efficiency

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d. Balance

e. factorial effects

Step 1: Setting up Hypothesis:

Null Hypothesis (H_0) = $\mu_A = \mu_B = \mu_C$

There is no significant difference in the sample means of different brands of batteries.

Alternative Hypothesis (H_1) = $\mu_A \neq \mu_B \neq \mu_C$

There is significant difference in the sample means of different brands of batteries.

Step 2: Test Statistics:

Under H_0, for one ANOVA, the test statistic is:

$$F_{cal} = \frac{MSC}{MSE}$$

$$MSC = \frac{SSC}{k-1}$$

$$MSE = \frac{SSE}{n-k}$$

S.N	Brand A (X_1)	Brand B (X_2)	Brand C (X_3)	(X_1^2)	(X_2^2)
1	40	60	60	1600	3600
2	30	40	50	900	1600
3	50	55	70	2500	3025
4	50	65	65	2500	4225
5	30	–	75	900	–
6	–	–	40	–	–
	$T_1 = \Sigma x_1 = 200$	$T_2 = \Sigma x_2 = 220$	$T_3 = \Sigma x_3 = 360$	$\Sigma x_1^2 = 8400$	$\Sigma x_2^2 =$

Grand Total $(T) = T_1 + T_2 + T_3$
$= 780$

Correction factor $(C.F) = \frac{T^2}{n}$
$= \frac{(780)^2}{15}$
$= 40560$

Total Sum of Squares $(TSS) = \Sigma x_1^2 + \Sigma x_2^2 + \Sigma x_3^2 - C.F$
$= 8400 + 12450 + 22450 - 40560$
$= 2740$

Sum of squares due to column $(SSC) =$
$\Sigma\left(\frac{T_j^2}{n_j}\right) - CF$

$= \frac{T_1^2}{n_1} + \frac{T_2^2}{n_2} + \frac{T_3^2}{n_3} - C.F$

$= \frac{200^2}{5} + \frac{220^2}{4} + \frac{360^2}{6} - 40560$

$= 8000 + 12100 + 21600 - 40560$

$= 1140$

Sum of squares due to Errors (SSE)
$= TSS - SSC$
$= 2740 - 1140$
$= 1600$

$(x_3)^2$
3600
2500
4900
4225
5625
1600
So $\Sigma x_3^2 = 22450$

One-way ANOVA Table:

Source of variation	Sum of Squares (SS)	Degree of freedom (D.F)	Mean Sum of Squares (MS)	F cal:
Between the samples (column)	SSC = 1140	$k-1$ = 3 - 1 = 2	$MSC = \frac{SSC}{k-1}$ = 570	$\frac{MSC}{MSE}$ = 4.27
Within the samples (error)	SSE = 1600	$n-k$ = 15 - 3 = 12	$MSE = \frac{SSE}{n-k}$ = 133.33	
Total	TSS = 2740	$n-1$ = 15 - 1 = 14		

Step 3 - Level of Significance (α):

Here, the level of significance is 0.05 or 5%.

$\alpha = 0.05$

Step 4: Critical value

Tabulated value of F at 5% level of significance for (2, 12) is:

$F_{tab} = 3.8853$

Step 5: Decision

Since $F_{cal} > F_{tab}$, H_0 is rejected and H_1 is accepted.

Step 6: Conclusion

There is no significant difference among sample means of different

brands of batteries.
The three brands have significantly differed average lifetime at the 5% level of significance.

Brand A:
Mean: $\bar{x}_A = \frac{200}{5} = 40$

Brand B
Mean: $\bar{x}_B = \frac{220}{4} = 55$

Brand C
Mean: $\bar{x}_C = \frac{360}{6} = 60$